MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 304TH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Triode

See figure 1

Mounting position: Vertical, base up or down

Weight: 9-ounces (255.15 grams) nominal

ABSOLUTE RATINGS: C Telegraphy

Parameter:	Fl	Ef	Eb	Ec	Ib	Ic	Pg	Pp
Unit:	MHz	V	Vdc	Vdc	mAdc	mAdc	W	W
Maximum:	8	5.25 or 10.5	2,000	-300	900	140	60	300
Minimum:	---	4.75 or 9.5	---	---	---	---	---	---
TEST CONDITIONS:	---	5.0 Vac	1,000	Adjust	300	---	---	---

GENERAL:

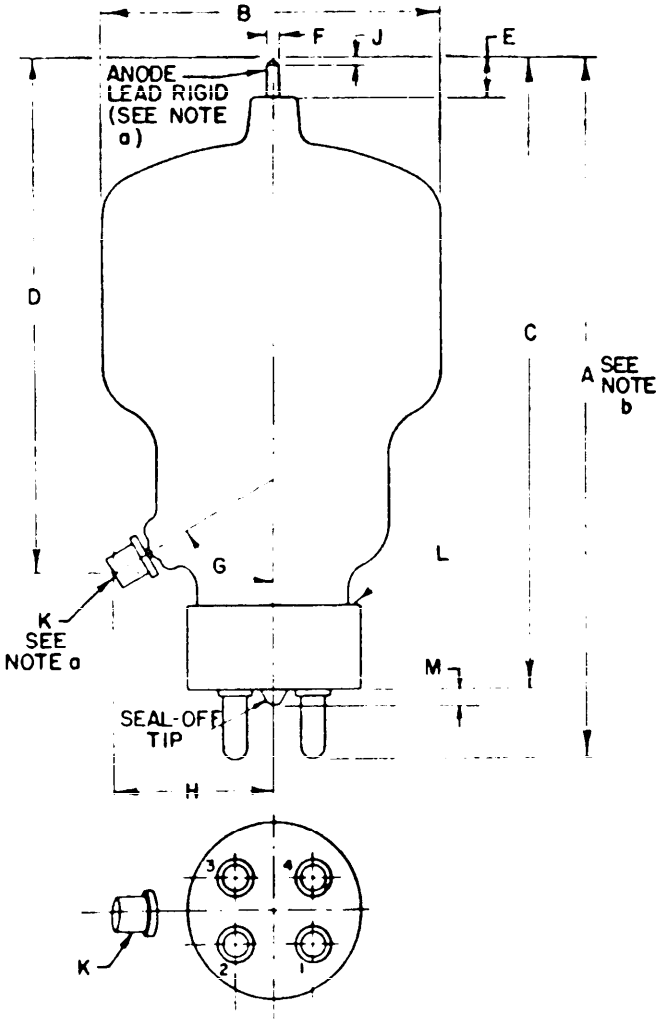
(E) Qualification - Not required.

(E) denotes changes

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Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 1	1					
1266	Total grid current	2		Ic	---	-25	µAdc
1231	Peak emission		eb = ec = 2,500 v	is	10.0	---	a
1261	Electrode voltage (1) (grid)			Ec(1)	-8.0	-18.0	Vdc
1261	Electrode voltage (2) (grid)		Eb = 14 kVdc; Ib = 1 mAdc	Ec(2)	---	-1,000	Vdc
1266	Primary grid emission		Ic = 325 mAdc; t = 15 seconds; Ef = 5.5 Vac; anode floating	Isg	---	-500	µAdc
1301	Filament current		Connect in parallel	If	24.0	28.0	Aac
	Quality conformance inspection, part 2						
1236	Power oscillation		Eb = 3,000 Vdc; Ec/Ib = 300 mAdc; F = 8 MHz; Rg = 10,000 ohms	Po	540	---	W (useful)
1316	Amplification factor			Mu	19.0	24.0	---
1331	Direct-interelectrode capacitance			Cgp	8	11	pF
				Cin	12	16	pF
				Cout	---	1	pF
	Quality conformance inspection, part 3						
1031	Low-frequency vibration	3	No voltage applied	---	---	---	---
1036	Bump	3	Angle = 10°	---	---	---	---
---	Life test		Group C; power oscillation; t = 500 hours	---	---	---	---
---	Life-test end points						
1231	Peak emission		eb = ec = 2,500 v	is	8.0	---	a
1266	Primary grid emission		Ic = 325 mAdc; t = 15 seconds; Ef = 5.5 Vac; anode floating	Isg	---	-1.0	mAdc

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PIN CONNECTIONS	
Pin No.	Element
1	f1
2	f1
3	f2
4	f2
top lead	a
cap	g

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	7.250 (184.15)	7.625 (193.68)
B	- - -	3.563 (90.50)
C	6.500 (165.10)	6.875 (174.63)
D	5.313 (134.95)	5.688 (144.48)
E	.438 (11.13)	- - -
H	1.688 (42.88)	1.813 (46.05)
Quality conformance inspection, part 3 (see note c)		
F	.120 (3.05)	.126 (3.20)
K	Cap: C1-23 (EIA)	
L	Base: A4-62 (EIA)	
M	- - -	.187 (4.75)
Reference dimensions		
G	60°	
J	.031 x 45° or .005R (0.79 x 45° or 0.13R)	

NOTES:
a. Heat radiating connectors HR-6 (grid) and HR-7 (anode) or equivalent, are recommended for all applications.
b. Dimension A includes .060 (1.52 mm) solder tolerance on length of pins.
c. These dimensions shall be checked annually with the following sampling plan:

n1 = 4 c1 = 0 where c2 represents the total allowable failures
n2 = 4 c2 = 1 for the first and second samples combined.

In case of failure after double sampling, the failing dimension(s) shall become quality conformance inspection (QCI), part 2, for three successful consecutive submissions, at which time the test may revert to the QCI, part 3 basis.

FIGURE 1. Outline drawing of electron tube type 304TH.

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NOTES:

1. The AQL (percent defective) for each test listed under QCI, part 1, shall be 0.65, inspection level II.

2. This test shall be the first test performed at the conclusion of the holding period.

3. This test shall be performed during the initial production and once each succeeding 12-calendar month period in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes having an acceptance number of zero, and a second sample of three tubes having a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, code level D. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 99
DLA - ES

User activities:
Army - AR
Navy - AS, CG, MC, OS
Air Force - 11

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5960-3349)

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MIL-E-1/884E	ELECTRON TUBE, POWER TYPE 3[illegible]

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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